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ZOO103 MID TERM SHORT NOTES BY 03460004097

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اَللّٰھُمَّ صَلِّ وَسَلِّمْ وَبَارِكْ عَلٰی سَیِّدِنَا وَمَوْلَانَامُحَمَّدٍ نِ النَّبِیِّ الْاُمِیِّ الْحَبِیْبِ ¤ الْعَالِیْ الْقَدْرِالْعَظِیْمِ الْجَآہِ وَعَلٰی آلِہ وَصَحْبِہِ وَبَارِكْ وَسَلِّمْ



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Muhammad Iqbal arain ♡ Nazia kanwal sahiba

MCQS:

TAXONOMY IS DERIVED FROM GREEK WORDS "**TAXIS**" WHICH MEANS ARRANGEMENTS AND "**NOMOS**" MEANS LAWS /METHODS

CAROLUS LINNAEUS, FATHER OF MODERN TAXONOMY.

ARISTOTLE-FATHER OF BIOLOGICAL CLASSIFICATION

SPECIES OF ROBINS IS NOT A **TAXON BUT CATEGORY**

اپنا لکھا مجھے معیوب لگے
وقت لکھے گا تعارف میرا

وہ ﷺ اتنا دیتے ہیں کہ جھولیاں بھر دیتےہیں
میرے آقا کے جودو سخا پہ لاکھوں درودوسلام

اقبال گروپ آف بی ایس زووالوجی

MUHAMMAD IQBAL ARAIN
DOCTOR NAZIA KANWAL

یہ کہتی ھے بلبلِ باغ جِناں نہیں کوئی رضا کی طرح سہربیاں
نہیں ہند میں واصف شاہ ہدیٰ مجھے شوخیِ طبع رضا کی قسم
(اعلٰحضرت امام احمد رضا خان بریلوی)
اقبال گروپ آف بی ایس زووالوجی

MUHAMMAD IQBAL ARAIN
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Qno1 Write the two important Contributions of Carolus Linnaeus to taxonomy ?

1-A hierarchical classification system

2- The system of binomial nomenclature

Qno2 Write The Three kingdoms proposed by Carolus Linnaeus?

• Animals

• Plants and

• Minerals

Qno3 What are kingdoms according to Carolus Linnaeus?

Carolus Linnaeus proposed that there are three broad groups called

Kingdoms into which whole nature could fit.

Qno4 What is classification?

It is a systematic method of arranging organisms into

different groups and subgroups, based on their similarities and their differences.

Qno5 Write the factors that causes classification?

Cell nature

• Cells • Mode of Nutrition

• Level of Body Organization

Qno6 Write the different characteristics of Aristotle to established collective catagenesis or genera?

1-Blooded and bloodless

2-Two-footed vs four footed

3- Hairy vs feathered

4-With or without an outer shell.

Qno7 What is systematics?

According to Simpson (1961) Scientific study of kinds and

diversity of organisms and of any or all relationship among them.

Qno8 Describe the Component fields of systematics?

• Component fields of systematics are:

• Biodiversity

• Taxonomy

• Classification

• Nomenclature

• Biogeography

• Evolutionary biology

Qno9 What is taxon?

Taxon

• According to Simpson (1961) taxon is a group of real organisms recognized

as formal unit at any level of hierarchic classification

• Blue birds, song birds, vertebrate refer to groups of organisms.

Qno10 Theoretical biology involve?

• Evolution

• Ecology

• Molecular Biology

• Evolutionary Biology

Qno 11 What is evolution?

Evolution is a process by which plants and animals

that exist today have been evolved from plants and animal of past.

Qno12 Which Contribution of Systematics involved in applied Biology?

Agriculture and forestry

• Biological control

• Public health

• Quarantine

• Wild life management

Mineral prospecting

• National defense

• Environmental problem

• Soil fertility

• In commerce

Qno13 Describe the role of systematics in commerce?

Systematics play role in increasing and improving the qualities of commercially used animal products such as:

• Honey

• Silk

• Lac

• Dyes etc

Qno14 What are capsid ?

Viruses are DNA and RNA shielded by protein coat called capsid.

Qno15 Write the name of five kingdoms?

Monera

• Protista

• Fungi

• Plantae

• Animalia

Qno16. Describe the Types of organisms?

1-prokaryotes – kingdom (Monera)

2-Eukaryotes -types (unicellular and multicellular)-multicellular without cell wall –(kingdom animals)

with cell wall -plantae(autotrophs) and Fungi (hetrotrophs)

Unicellular -kingdom (protesta)

Qno17 What is taxonomy?

Taxonomy is the science of naming, describing and classifying organisms including all plants, animals and microorganisms of the world.

Qno18 What is the downward classification?

Downward classification are the principles of logical division which consist of

dividing the larger groups based on dichotomy into smaller groups. e.g,Animals.

Qno19 What is taxonomic explosion ?

By 1955 taxonomy got a new label of biological landmark and reach such a status that it was called taxonomic explosion.

Qno20 Which terms are applicable to taxonomic study?

• Instead of morphology as an original base, phylogenetic adaptations, embryological patterns, biochemical variations, genetic variabilities and similarities, behavioral characteristics are all applicable to taxonomic study.

Qno21 What are siblings species?

Sibling Species: Two or more than two closely related species which are morphologically alike but behaviorally or reproductively isolated from each other.Examples are Drosophila persimilis and D. pseudoobscura.

Qno22 What are Monotypic Species?

species with a single subspecies, called monotypic species

Example: Bardala labarda.

Qno23 What are Polytypic Species?

• A species contains two or more subspecies called polytypic species.

• Examples are tiger, Panthera tigris which has several subspecies.

Qno24 What are endemic species?

• The species which are found in a particular region, called endemic species.

Example:

• The Darwinian finches are the endemic species of Galapagos Islands

• Indus dolphin in Pakistan.

Qno25 Define Deme?

A deme is a sub-population that can freely interbreed.

Qno26 What is Cline?

• A cline is the gradual change in the phenotypes and genotypes of a species across a geographical gradient.

Qno27 Describe the Types of cline?

• Smooth cline

• Step cline

Qno28 What is the clade ?

• A clade is a group of organisms that evolved from a common ancestor.

E.g Eutheria is one mammalian clade .

Qno29 What is speciation?

The creation a new species.from the old species is called speciation.

Qno30 Describe the Anagenesis and Cladogenes?

1- **Anagenesis**
2- **C**ladogenes

Anagenesis: A single species may change over time into a new form that is different enough to be considered a new species.

Cladogenesis:

• A species may become split into two groups that no longer share the same gene pool.

Qno31 Write the Cusses of speciation?

• Geographical Isolation

• Behavioural isolation

• Reproductive isolation

Qno32 Describe the Types of speciation?

• Allopatric speciation

• Peripatric speciation

• Sympatric speciation

• Parapatric speciation

Qno33 What is Allopatric Speciation?

• The most common form of speciation occurs when populations of a species become

geographically isolated.

.Qno34 What is allopatric Speciation?

•When populations become separated, gene flow between them ceases over time, the populations

may become genetically different is called allopatric speciation.speciatio

Example : Speciation is the Galápagos finch

Qno35 What is Sympatric Speciation?

A speciation in which a species is evolved from a single common ancestral species while inhabiting the

same geographical range.

E.g More common in plants.

Qno36 What is Popyploidy in plants?

• Plants produce offspring that are polyploid. Hence, the offspring live in the same environment as their parents but are reproductively isolated..

Qno37 What is Parapatric speciation?

• It occurs when populations are separated not by a geographical barrier, such as a body of water,

but by an extreme change in habitat.

Qno38 What is Paripatric Speciation?

• Peripatric speciation is a mode of speciation in which a new species is formed from an isolated peripheral population.

Example

• An example of this is the London Underground mosquito, a variant of the mosquito Culex pipiens,

which entered in the London Undgerground in 19th century. • It is found in the London Underground railway system

Example

• Evidence for its speciation includes genetic divergence, behavioral differences, and difficulty in mating.

Qno39 Describe Aristotle and Plato concepts about Typological species?

“A certain pattern of features is found in each species of organisms and all members of species show maximum equality in this format”

Example: certain pattern in dogs.

Qno40 Describe the Types of Typological species concept ?

• Monotypic species

• Polytypic species

Qno41 Describe Problems with Typological species concept ?

• Polymorphism within populations.

• Sibling or cryptic species.

• Geographical variations among population

Qno42 Why members of species resemble ?

When organisms breed they shuffle their gene pool which gives the species its identity.

• Genes are not shared with other species so they look different.

Qno43 Write the constitute of Biological species concept?

Ecological unit

Reproductive unit

Genetic unit

Qno44 Why Dobzhansky (1951) and Hanson (1981) accepted the definition of biological Species concept?

They accepted because of two reasons:

• Gene pool

• Reproductive isolation

Qno45 What Decedents are called in uniparental or asexual reproduction?

Decedents are called agamospecies, binoms or paraspecies.

Qno46 What is evolutionary species?

An evolutionary species “is a single lineage of ancestor-descendant populations of

organisms which maintains its identity from other.

Qno47 Describe Taxonomic Characters ?

Morphological character

• Physiological character

• Behavioral character

• Ecological character

• Geographic character

• Molecular and biochemistry character

• Ontogenetic character

Qno48 Describe morphological character?

External morphology

Genetelic structure

Internal morphology

Embryology

Karyology and other cytological differences

Qno49 What is karyology?

Karyology is description of size, shape and number of chromosomes.

What and which are physiological characters?

Physiological characters are low weight characters and are variable.

There are following characters:

• Metabolic activities

• Body secretion

• Serological protein and other biochemical differences .

• Genic sterility factor

Qno 50 What are the antigens?

Antigens are the substances capable for inducing the formation of antibodies when

introduced into the blood stream of animals.

Qno51 What are Molecular and biochemistry character ?

• Immunological distances

• Electrophoretic differences

• Amino acid sequence of protein

• DNA hybridization

• DNA and RNA sequence

Qno52 Write the DNA hybridization

• This method is used to determine the genetic distance between two organisms

• Mainly measure the genetic similarity.

Qno53 What is weighting?

Process to determine that which character is more reliable is called weighting.

Qno54 Describe the models of weighting?

1. Priori (before/without phyletic information)

Proposed by Artistotle

2. Posteriori

Proposed by Adenson

Qno 55 Describe Correlated Characters?

• Phyletic correlation

• High weight

• Higher genetic contents

• Diagnostic characters of Deuterostomes and Chordates

Qno56 Describe High weight characters ?

• Complexity

• Constancy

• Consistency

• Not affected by ecological shift

• Apomorphy

• Not a specific adaptation

• e.g. Morphological and behavioral character.

Q57 What is consistency?

• Character which is constant in a large group of species than variable character (Darwin 1859) called consistency.

OR

• A character which is consistently present in one group and consistently absent in related

groups should have high weight.

Qno58 Describe Low weight characters ?

• Not constant

• Not consistent

• Change with ecology

• Less complex

e.g. Physiological characte

Qno59 Describe Types of Non-Genetic variations ?

• Individual variations in Time

• Social Variations

• Ecological Variations

• Traumatic variation

• Post-morten variations

Qno60 Describe Ecological Variations?

A. Habitat Variation

B. Variations induced by climatic conditions

C. Host determined variations

D. Density dependent variations

E. Allometric variations

F. Neurogenic color variations

Qno61 What is Neurogenic variation?

• It is colour change in response to environment.

Qno62 Define Teratology?

The study of abnormalities of physical development.

Qno63 what is Taratogen?

Taratogen are the toxic substances which cause abnormal development.

E.g• Mercury, alcohol, lead, thalidomide

• Teratological variations are common in animals.

Qno64 Describe genetic variation?

• Evolution

• Natural selection

Causes of evolution :

• Mutations

• Genetic recombination

• Genetic drift

• Migration

Qno65 Describe major causes of variability in genetic variation?

• Mutations

• Genetic recombination

• Mutations are not always bad

(Sickle cell disease—present Malaria)

Qno66 What is polymorphism or discontinuous. Variations?

Certain members of population have certain conspicuous characters. Such discontinuous individual variations is called polymorphism.e.g **lady beetles**

Qno67 What is systematics collection?

Systematic collection

Systematic collection is a method of collecting valuable biological specimens and preserving them along with specimen documentations such as labels and notations. It possess various attributes:

• Raw material of classification

• Document earth's biodiversity

Qno68 What is adequate?

• More material is needed in s species with strong individual and geographical variation than in uniform species.

Qno69 What are the Objective of study of Systematic Collection?

Simple study or genetic variability

• When population analysis is attempted dealing with size, proportions, coloration, or polymorphism large samples from numerous locations are needed.

• For example in fishes where each population of a species may vary in some metric characters (finray,vertebrae), a single specimen per locality would tell us little.

Qno70 Describe Preservation of collection samples ?

The collected samples are preserved by;

• Wet method and

• Dry method

Qno71 Explain Wet and dry method?

Wet method

• Wet method of preservation involves the use of liquid preservatives to prevent deterioration.

• Chemicals like alcohols, formalin etc. are commonly used for specimen preservation. Animals

belonging to phylum porifera, annelida, nematodaare mostly preserved by wet method.

Dry method

• In this method, specimens are preserved in dry state is a traditional method to place

several kinds of insects in a box for an indefinite period allowing them to become dry.

• Not advisable for soft bodied insects.

• The process of pinning insect specimens is a dry method to preserve and display collections.

Qno72 Describe Collection methods used to collect specimen?

There are various collection

methods to collect specimens

• Black light trap

• Sweep net

• Suction device

• De-Vac

• Pit fall method

• Aspirator

• Beating sheet

• Trawling

• Dipper

• Hand picking

Qno73 What are draggers ?

The boats that are used for trawling are called trawlers or draggers.

Qno74 What is Curation and curator?

process of identification and organization of animal collection is called curation.

Curator:

curator is a person who manages, administers or organizes a collection for exhibition.

Qno75 What is Cataloging ?

• Museum number

• Original field number

• Scientific Name (at least generic name)

• Exact locality

• Date of collection

• Name of Collector

• Remarks

• Card systems/electronic data processing

.Exchange of materials

Qno 76 Describe the Important Properties of Texa Names?

Three important properties

• Uniqueness

• Universality

• Stability

Qno77 What is type specimen?

type is a specimen (or a group of specimens) which serves to illustrate the defining characters of a species or genus..

Qno78 Describe kinds of type specimen?

Holotype.

Paratype

Allotype

Syntype

Neotype

Monotype

Qno79 What is topotype and Topotypical population?

Specimens collected at type locality are called **Topotypes**.

• Population that occur at type locality is called **Topotypical population.**

Qno80 What is Nomen nudum ?

• A name published without satisfying the condition of availability is called nomen nudum..

Qno81 Write Types of Synonyms ?

Objective synonym

Subjective synonym

Qno82 Describe the abbreviation of following ICZC, (IUBS), ICZN?

ICZC: International Code for Zoological Nomenclature

(IUBS), International Union of Biological Sciences.

ICZN: (International Commission on Zoological Nomenclature)

Qno83 Describe the parts of ICZC?

The code proper

Appendices

Glossary

Qno84 Explain Appendices Glossary?

Appendices

• Appendices are like recommendations not to be followed strictly as article. These are five in number and are designated as A-E.

Glossary The terms used in the text are clearly defined in the Glossary.

Qno85 Describe the Applications of molecular systematics?

Solving taxonomic problems

Reorganization of tree of life

Identification of disease vectors

Testing food authenticity

Monitoring of illegal trade

Monitoring quality of water

Identification of damage specimens

Identification of sibling species and sub-species

Qno90 Describe PCR based techniques ?

• RAPD (Random Amplified

polymorphic DNA)

• ISSR (Inter-Simple Sequence Repeat)

• SSR (Simple Sequence Repeat)

• AFLP (Amplified Fragment length Polymorphism)

• EST (Expressed sequence Tags

• SCoT (Stop Codon targeted marker)

Qno91 Describe Advantages of Molecular Markers ?

• Ubiquitous

• Stably inherited

• Multiple alleles for each marker

• Devoid of pleiotrpic effects

• Detectable in all tissues, all ages

• Long shelf life of DNA samples

Qno92 Write the Application of Molecular Markers ?

• Fingerprinting

• Diversity studies

• Genetic maps

• Gene tagging

• F1 identification

• Seed testing

• Comparative maps

Qno93 What is an Arthropod?

• Any member of Phylum Arthropoda having joint legs are called arthropods.

E.g lobster, prawn, mites, insects, spiders, centipedes, millipedes etc.

Qno94 What is ecdysis?

Molting of the cuticle is called ecdysis.

Qno95 Why Appendages in arthropods are modified?

Appendages in arthropods are modified for a number of functions:

• Walking legs

• Mouth parts

• Antennae

• Claws

• Organs for mating

• Holding eggs

Qno96 Describe Four lineages of arthropods ?

• Trilobites – extinct

• Chelicerates – horseshoe crabs, spiders

• Uniramians – centipedes, millipedes, insects

• Crustaceans – crabs, lobsters, barnacles

Qno97 What is Dichotomous keys?

• A dichotomous key is a device for identifying unknown organisms to some taxonomic levels such as species, genus or family level.

Qno98 What is couplets?

Dichotomous keys Consists of series of paired statements, called couplets.

Qno99 What is lead?

Each statements in a couplet is called a **lead**.

Qno100 Define Catalogue ?

• Catalogue is mainly an index to the texa arrange in such a way that as to give a vivid picture of references for both zoological and nomenclatural purpose.

Qno101 Describe the Types of keys?

• Indented Key

• Bracket Key

• Pictoral Key

• Polyclave Key

Qno102 Describe the leg parts of spider ?

Parts of Legs :

• Coxa

• Traochanter

• Femur

• Patella

• Tibia

• Metatarsus

• Tarsus

• Claws

Qno103 Describe Parts of Venom gland and use ?

• Poison gland

• Poison Duct

• Fang

• Absent in the family Uloboridae. The glands are located either in the chelicerae or under the

Carapace.They are used to capture prey and defense purpose.

Qno104 Discuss Parts of Silk gland and use ?

• For capturing prey

• For food storage

• Travelling

• Ballooning

Qno105 Explain Reproduction in Spiders ?

• Male find a female

• A male spider deposits his sperm into a sperm web, holding it in his palps.

• Female spiders produce either one egg sac containing several to a thousand eggs or

several egg sacs each with successively fewer eggs. Females of many species die after producing the last egg sac.

Mcqs

Elton (1947) acknowledged the progress in ecology depends upon **accurate identifications of species.**

Molecular biologist use **genetic data** for establishing relationships among different organism.

DNA barcoding and taxonomist is the example of **molecular biology.**

Tamarixia radiata, a parasitic wasp was introduced from **Pakistan to California** to control this pest.

The scientific name of spider  is **Oxyopes javanus.**

Hottentotta tamulus, the **Indian red scorpion** also known as the **eastern Indian scorpion**, is a species of scorpion of the family **Buthidae**.

During World War II (**On November 3, 1944),** Japan released balloon bombs, into the **Pacific jet stream**

Balloons covered with sand contained a large number of **shells of micro-organisms.**

Phytoplankton and zooplankton are sensitive to changes in water quality.

organisms that enhance soil fertility are **bacteria,earthworms** etc.

A single domain can contain **one or more** kingdoms. Introduced by **Charl Woeseet al. In 1990.**

E- Chatton in 1937 introduced the term "**Procariotique**" to describe bacteria and "**Eucariotique**" to describe animals and plant cells.

Five kingdom classification was proposed by **Robert Whittaker in 1967.**

Simpson in 1961 gives a valuable survey of **History of Taxonomy**

During the 28 years of frederick smith activity he describe **25 genera, 702 species and subspecie**s.

Frederick Smith worked in the zoology department of **British Museum from 1849**.

• He published from **1851 to 1870** on **ant taxonomy**.

Thomas E.Mittler worked on history of higher taxonomy and also on the **classification of insects**

Mayr described the History of Taxonomy into **four periods.**

Linnaeus adopted the principle of downward classification in **15th** edition of his **book (Systema naturae)**

He for the first time applied **binomial nomenclature** to animal kingdom

Anderson 1763 (Botanist) applied this design for the **classification of plants.**

Cuvier recognized **five phyla**

Darwin believed in **fixity** of species but after observing **natural fauna in Galapagos island**, he begin to believe in the **plasticity** of species.

The phylogenetic tree of **Ernest Haeckel** also stimulated the empirical workers.

The greatest development of modern taxonomy started around **1930**.

The publication of the book '**new systematics'Become** a **landmark** in the History of Taxonomy.

Enough gene flow occurs between **demes** ensuring that all of it's members remain the same species.

Survey of taxonomic literature shows that there are more than **20 species** concepts.

Characters are usually described in terms of **character states.**

Protein antigens are **conservative hereditary** trait.

Only closely related species can **hybridize** successfully.

Some behaviours are seasonal and change within **24 hours**

Each species has **specific niche**

Structural changes can occur in host after **parasitic attack**

Characters which are functionally correlated have very **low weight**

According to Farris (1969) **Low phenotypic variability** be given **high weight**

Freshly molted bird of October is covered with **white spots and all feathers** show **whitish margins**

In **winter edges** of feathers **wear off.**

• In spring (beginning of breeding season) whole bird is **glossy black** without the mold of single feather.

Seasonal variations of consecutive generations (Seasonal plasticity) these variations called **polyhenism**

. African butterfly **Bicyclus anynana**

Honey bee 🦋 **female** May be from **10000 to 50,000** per colony

🐇 Ant queen Long life in some species **upto 30 years.**

When newly hatched nymphs of locust are reared under crowded conditions----**Gregarious phase**

• Less crowded condition---**Transitional phase**

• Separate rearing---**Solitary phase**

• These phase differ in anatomy, colour and behavioral characteristics----**distinct species**

Chameleons have **Chromatophores** in **three layers** below their **transparent skin.**

Xenthophors **(Yellow)** and erythrophores **(red)** in upper layer.

Iridophores or guanophores **(blue or white)** 2nd layer.

Melanophores **(dark colour)** in deep layer.

Chameleons change their colours in response to **temperature and to chamouflage.**

It is impossible to prevent **post-mortem** changes in some animals.

Deep yellow plumage of **twelve wire** bird of **paradise----white**

Chinese jay plumage changes from **green to blue** in collection.

Wasps over-expressed to **cyanide** turn **bright red.**

Many of the variants in a population are sex-associated. **They may be sex limited (express themselves in one sex only)**

When two sexes are similar morphologically—**PSD helpful**

Celebrated case of King parrot, **Eclectus roratus** (Muller)

King parrot Eclectus roratus Male is **green with orange bill**

Female is **red and blue with black bill**

. Males of African Ant Dorylus are so **unlike** others ants that they were not recognized as ants and

considered belong to different family.

Individuals that show male characters in one part and female characters in other parts of body are called **Gynandromorphs**

Blending of male female char**ormalin/alcohol Detergent drops**acters is called **Intersexes Example: Gipsy moth, Lymantaria**

In **cynips (gall wasp**) agemic generation is different from **bisexual** one.

• In **aphids parthenogenetic** wingless females are different from winged females of **sexual generation.**

In some species of **slugs (Arion)** several several strains are found which differ in their **sexuality**.

Sexually reproducing strains are highly **polymorphic** in their enzymes

self fertilizing hermaphroditic strains are highly **homozygous**

. Species of the **snail genus Melani**a **(fresh water and brackish water)** have been described largely on the basis of **shell characters** as presence or absence of **spines and diagonal** versus spiral ribs.

Entomologists often collect night flying insects using a **black light and sheet.**

Sweep nets are used to collect **arthropods** from **long grass** and other types of vegetation.

Suction device/De-Vac Used for collection of arthropods from **vegetation, long grasses, forest, orchards** etc. • **Engine operated**. • Large scale insect collection.

Pit Fall Method Filled with Filled with **formalin/alcohol Detergent drops to** beak surface tension to beak surface tension.

Small insects or insects in hard to reach places, can be collected using an **aspirator**

. • To study insects that live on branches and leaves, like **caterpillars**, a **beating sheet** is a tool to use.

Trawling is a method of **fishing** that involves pulling a fishing net through the water behind one or more boats.

The most commonly used larval collection method is the "**standard dipper.**

American Museum of Natural History (AMNH), **New York**

Type designation should be **clear** and **unambiguous**.

Priority starts from **May 01, 1753** ,Apply only to **family level** and not according to **Zoological code**

Conservation can may be avoided in Zoology as these names fall in formal category of **Nomen oblitum.**

• In taxonomy, a forgotten name; i.e. the name of a senior synonym that has not been used

in the zoological literature for at least **50 years**. • Such names should not be used if they upset a better-known name, but should preferably be suppressed by the **ICZN**

. • If a current name for a taxon have archaic or obscure **homonym (same** name to different taxon), the current name can be declared as **Nomen protectum (available name).**

The first Scientific name applied to the **leopard shark** was **Triakis californica.** Proposed by British Zoologist **Edward Gray (1851).**

• In 1855 John Edward Gray published the name of **Pronghorn** (Antilocarpa anteflexa) Based on **pairs of horns**

• Taxonomic name which should be used is called **Valid (Accepted, correct) name.** The name must have been published according to requirements detailed in **Articles 8-20.**

Article 11-19 describe **prerequisite** for making a specific **name available. Particularly article 13a is very important.**

The term nomenclature comes from **Latin words**Nomen which means Names • Calare mean to call • Call by names.

Phylogenetic and Phylogeographic studies started with the introduction of **mDNA** markers in population genetic studies.

First phylogenetic tree based on molecular data **1967 Fitch and Margoliash**

Type I: Associated with genes of **known function**. **Alloenzymes** are type I markers as the **protein**

they encode are associated with genes of known function.

Type II: Associated with genes of **unknown function**. **Microsatellite** and other neutral markers are type II markers.

alloenzyme coded by different allele present at same locus.

Isoenzymes which are coded by genes located at **different loci**

Mitochondrial DNA Markers **Non-Nuclear DNA ,Homologous markers.**

Microsatellite having 2-6 nucleotides and Microsatellite sequences are composed of **non-coding DNA**

and are **not parts of genes.**

SNPs **Single Nucleotide substitution**

Most abundant **polymorphism** in **genome** of an organism Can be detected by **PCR, microchip**

array or fluorescence technology.

DNA barcoding have CO1 gene 658 base pairs High mutation rate Widely used in identification of arthropods

Protein (Biochemical markers) Available since **1950s** Can be studied using **Gel electrophoresis** **e.g esterase, Acid phosphatase, peroxidase etc.**

RFLPs (Restriction Fragment Length polymorphism) Analysis Introduced by **Bostein et al. 1980**

Hybridization based techniques are also called **Non-PCR**

Glycerin in alcohol help to **preserve colours** and to keep **specimens flexible.**

70-75% alcohol is used for long term storage

95% alcohol is used for of **DNA studies**

Iropropyl Alcohol Available strengths **(90%, 70%)** ,For use dilute **upto 40%**

Coelenterates are difficult to preserve ,**Coelenterates** Preserve in **70% alcohol or 5% formalin**

In **arthropods** Larger invertebrate need to be injected **10% formalin** to prevent from **rotting**

Arthropods have **Seven million species**

Nearly **one million** species identified so far, mostly insects.

The exoskeleton, or cuticle of arthropods is composed of **protein and chitin.**

In arthropods **Aquatic members- gills for gas exchange;** terrestrial members-**tracheal system** of branched tubes leading from surface throughout body

Striated muscle associated with the exoskeleton, and **smooth muscle** associated with the gut

In arthropods From **4 lineages arose five major** classes of arthropods.

• If at each step there are only two alternative, the key is said to be

DICHOTOMOUS (sequential key, analytical key, pathway key) **POLYTOMUS** key

These most commonly published keys were first published by **Jean B. Lamark 1778**

The Catalogue currently compiles data from **168 peer-reviewed** taxonomic databases**.**

The Catalogue interface is available in **twelve languages** and is used by research scientists,

citizen scientists, educators, and policy makers.

Kingdom arthropods have More than **46000** species Ranked **7th** in animal Kingdom

Spider 🕷 has **Four pairs of legs** Two body and No antennae.

Spiders have **two** body segments Front segment is called **Cephalothorax**

Spider have **Silk glands** and **Claws 2-3** at the end of legs

Most spiders have **eight eyes.** Some species have **six or fewer eyes,** but they always come in an **even** number.

• Some species of spiders, such as those that live in **caves or under the soil**, have **no eyes** at all. Even

those species with **eight eyes** don't usually see **very well.**

99% of all spiders have **8 eyes**

• **1%** nearly all have **6**, but there are a few exceptions.

Spiders, in fact, do taste, and also smell, through special **sensory organs** on their **legs**, as well as on

their **pedipalps**.

Epigyne or epigynum is the external genital structure of female spiders.

Completed mid term lecture no 1-95

Current papers 2022

Qno1 What are pedipalps and epigyne ?

Padipalp/Epigyne

• **Pedipalps** (commonly shortened to palps or palpi) are the second pair of appendages of chelicerates

• **Epigyne or epigynum** is the external genital structure of female spiders.

Qno2 What is prosoma?

The anterior body part of the jumping spider is the prosoma orcephalothorax.

• It is covered above by a shell or carapace:

Qno3 What Are the ecological variations? 5 marks

A. Habitat Variation

B. Variations induced by climatic conditions

C. Host determined variations

D. Density dependent variations

E. Allometric variations

F. Neurogenic color variations

A. Habitat variation

• Populations of single species in different habitat are visibly different.

Oyster in still water

• Round or broader

Oyster in strong currents

• Narrow, elongated and usually straight

Oyster in clean water on pebble

• Lower valve is deep and more or less sharply ribbed to acquire strength

• Not needed attachment is with perfectly flat surface

Oyster in muddy area

• Valves are less commonly ribbed.

B. Variations induced by climatic conditions

• Animals with phenotypic plasticity

• Variations with temperature, draught, cold etc.

C. Host determined variations

• Braconid wasp, Apanteles flaviconchae spin white cocoons if reared from blue

green caterpillars of Choliase philodice

• Golden cocoons if reared from yellow green caterpillars of same species.

D. Density-Dependent variations

• When newly hatched nymphs of locust are reared under crowded conditions called **Gregarious phase**

• Less crowded condition called **Transitional phase**

• Separate rearing called **Solitary phase**

• These phase differ in anatomy, colour and behavioral characteristics.

Allometric variation

• Allometry results in misappropriate size of some structure in relation to rest of the body.

• This results in animals of different sizes----Allometricvariabilit

Neurogenic variation

• It is colour change in response to environment.

•Chameleons have Chromatophores in three layers below their transparent skin.

• Xenthophors (Yellow) and erythrophores (red) in upper layer

• Iridophores or guanophores (blue or white) 2nd layer

• Melanophores (dark colour) in deep layer.

Chameleons change their colours in response to temperature and to chamouflage.

Qno4 Describe evolutionary species concept? 3 marks

Evolutionary species concept

• Simpson (1940s)

• An evolutionary species “is a single lineage of ancestor-descendant populations of organisms which maintains its:

1. Identity from other such lineages [in space and time]

• Own evolutionary tendencies• historical fate” (Wiley, 1981)

Qno5 Describe probability sampling types? 5 marks

Probability Sampling are:

• Random Sampling

• Systematic Sampling

• Stratified sampling

• Cluster Sampling

• Multistage Sampling

Random Sampling: If every individual of target population has an equal chance of selection then it will be called random sampling.Simple random sampling is the easiest form of probability sampling. No biasness and expert opinion is required in this method.

Systematic Sampling: In this sampling method first sample from the population is taken randomly but after that samples are selected systematically at a regular interval. Examples:

Shopping Centre

Stratified sampling: Stratified sampling is a probability sampling technique wherein the researcher divides the entire population into different subgroups or strata, then randomly selects the final subjects proportionally from the different strata.

Cluster Sampling: Cluster sampling is the sampling method where different groups within a population are used as a sample

Multistage Sampling: Multistage sampling is a method of obtaining a sample from a population by splitting a population into smaller and smaller groups and taking samples of individuals from the smallest resulting groups.

Qno6 Social variations in ant? Long and short

Social variation in ants

• Ants live in complex social colonies.

• **Queen**: founder and leader of the colony.

• Primary goal is to populate colony .

• Lay thousands of eggs

• Queen chambers are deep within the colony for protection.

• Long life in some species upto 30 years.

Drones: Only function is to fertilize female

• Die after fertilization

• Rarely encountered outside the colony

• **Workers**: Foraging and protection.

• Sometime: queen lay eggs, winged male and females-Alates or reproductive.

After maturation leave the colony in swarms.

• Establish new colony

Qno7 What are co-related characters?

A particular gene will often have effect on several different traits process known as

ploetrophy. Because several traits often share part of the same underlying genetic architecture, they are

genetically known as correlated characters.

Characters which are functionally correlated have very low

weight • They have low phylogenetic content • Do not deserve to be treated as separate character

• Analogues organs birds wings, bats and insects (analogus)/convergent evolution.

Qno8 What are the feeding habit in jumping spider ?

They are mostly carnivorous. Jumping spiders eat insects like flies, moths, and grasshoppers. They are predominantly carnivores, though they have been known to eat nectar. In general, jumping spiders eat anything they can get their chelicerae (jaws) around.

Qno9 Describe five kingdoms classification?

Two Kingdom classification was proposed by Robert Whittaker in 1967.

Morgulis and Schwatz in 1988 modified R. Whittaker classification.

• They considered genetics along with cellular organization and mode of nutrition in classification

. Five kingdoms proposed by

Robert Whittaker include:

• Monera

• Protista

• Fungi

• Plantae

• Animalia

Qno10 Describe coutship behavior in lynx spider?

In a typical mating sequence of the green lynx spider, courtship lasted 11 minutes. Mating took 10 minutes, consisting of several copulations measured from the first approach of a male palpus to the female's epigynum until the final separation. During courtship "dance“ colored or iridescent parts of the body are displayed. • In addition to the display of colors, jumping spiders perform complex sliding, vibrational, or zigzag movements to attract females.

Qno11 What is three domain system?

1. Archaea

2. Eubacteria

3. Eukaryota

A domain is a broadest taxonomic category.

• All organism belong to one of the three domains, depending upon their characteristics.

• A single domain can contain one or more kingdoms.

• Introduced by Charl Woeseet al. In 1990.

Qno12 non probability sampling?

Non-Probability Sampling are:

• Purposive sampling

• Quota Sampling

Qno12 What is internal morphology?

Internal morphology

• Internal anatomy provides Taxonomic Characters for all groups of higher animals.

• The extend to use vary from group to group.

• The skull of routinely used in classification of higher mammals.

• Teeth and bones of fossils and other hard parts

Qno14 problem with Typological species concept?

• Polymorphism within populations.

• Sibling or cryptic species.

• Geographical variations among populations

Qno15 Describe similarities and differences of eubacteria and archaebacteria?

Eubacteria :

Cell type : prokaryotic cell

Cell wall :contains peptidoglycan

Introns: no introns

Ribosomal protein:no ribosomal protein

Disease : can be parasitic cause disease

Gene resemblance: No resemblance

Archaea :

Cell type: prokaryotic

Cell wall: carbohydrates and protein or no cell wall

Introns: introns present

Ribosomal protein: ribosomal protein present

Disease: none are parasitic

Gene resemblance : eukaryotes

Qno16 :Define weighting character : Process to determine that which character is more reliable is

called weighting. 1. Priori (before/without phyletic information) • Proposed by Artistotle 2. Posteriori • Proposed by Adenson (after phyletic information collection) • All classification is based on this mode.

Qno17 morphology and it's characters ?

Morphology of animal: morphology, in biology, the study of the size, shape, and structure of animals, plants, and microorganisms and of the relationships of their constituent parts.

Morphological characters include: • External morphology• Genetelic structure• Internal morphology• Embryology• Karyology and other cytological differences.

Qno18 Describe jumping spider behavior?

Jumping spider behavior: Jumping spiders are generally diurnal, active hunter. There well developed internal hydrolic system extend their limbs by altering the pressure of their body fluid (hemolymph) within them. This enables the spider to jump without having large muscular legs like a grasshopper.

Qno19 Describe sampling method?

1-Probability Sampling

Types of sampling:

Probability Sampling

• Random Sampling

• Systematic Sampling

• Stratified sampling

• Cluster Sampling

• Multistage Sampling

2-Non-Probability Sampling

Types of non- sampling method

• Purposive sampling

• Quota Sampling

Qno20 Describe social variations of honey bee?

In social insects there are definite groups of individuals in a colony.

• Each type of individual in a colony perform specific function.

• For examples in a colony of Honey bee:

• Queen, males (drones) and workers, soldiers Social Variations.

• workers: All workers are females, product of fertilized eggs.

• They are in majority in the colony.

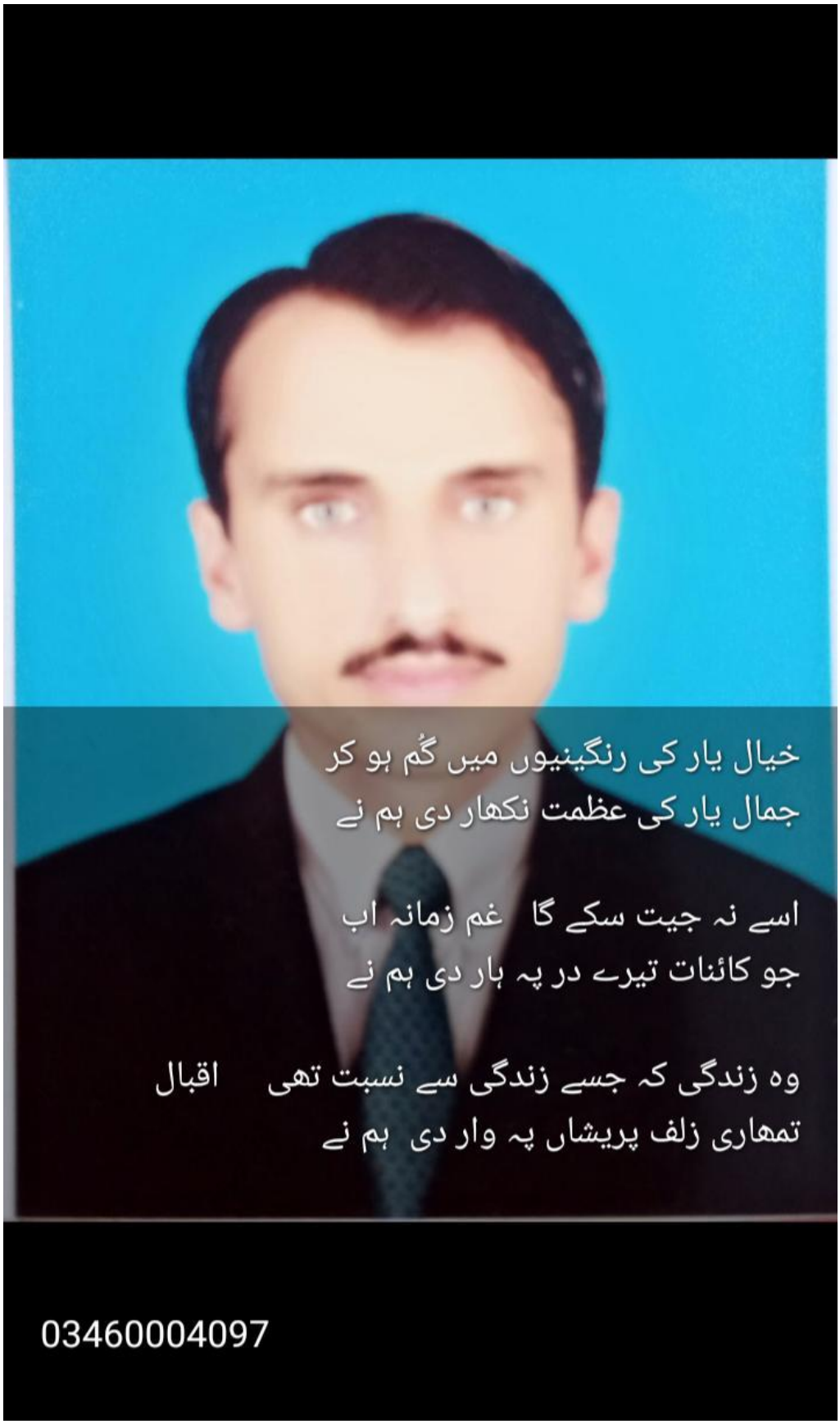
• May be from 10000 to 50,000 per colony.

• development.

• Projection of colony, food collection, feeding of queen, take care of young

ones, cleaning of hive etc.

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نہیں کوئی حسرت شہرہ آفاق ہونے کی اقبال

بس اتنی سی تمنا ہے تیرے نام کے ساتھ امر ہو جاؤں

وہ چند لمحے جو گزرے تیری رفاقت میں

اقبال

نہ جانے جانِ جاناں میرے کتنے برس کام آئے

اقبال گروپ آف بی ایس زووالوجی

شاہین گروپ آف بی ایس زووالوجی

یا اللہ پاک ہمارے آقا کریم ﷺ کے صدقے محمد اقبال آرائیں کو بخش دے اور انکی دین کی راہ پہ چلتے ہوئے مغفرت فرما دے آمین

اللہ حافظ دعا میں یاد رکھنا